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TO WHOM IT MAY CONCERN

Letter of Intent: Master's Thesis Project

This is to confirm that Ms. Chansa Raingam (Roll No. 254154003) is currently conducting her Master's Thesis Project under my supervision within our research group at School of Agro and Rural technology, IIT Guwahati. Her thesis topic is titled "Integrating PlanetScope Data and Dual-Pol Radar Vegetation Index for Crop Biomass Monitoring."

Her ongoing work is integral to our current project workflows, focusing on high-resolution optical and microwave remote sensing integration. Throughout the tenure of her thesis project, she is actively involved in the following key technical workflows including but not limited to:

Ground Data Collection: Coordinating and executing field campaigns to acquire ground truth observations, including biophysical crop parameters, precise location tagging, and ground-level biomass measurement protocols.

Satellite Data Processing: Preprocessing high-spatial-resolution PlanetScope optical imagery and Dual-Polarimetric Synthetic Aperture Radar (SAR) datasets. This includes polarimetric processing, radiometric calibration, terrain correction, and calculating Radar Vegetation Indices (RVI).

Data Analytics & Integration: Developing analytical models to fuse high-cadence optical imagery with dual-pol radar parameters for robust crop biomass estimation, statistical validation, and temporal monitoring.

Ms. Chansa has demonstrated dedication and strong analytical potential in executing these research tasks. This letter is issued upon her request for official academic and professional reference purposes.

Date: 31-08-2026

Place: Guwahati, India


With regards,

Dr. Dipankar Mandal
Assistant Professor
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